

Session 7: Strings (C Version)

Topics Covered 1. String Theory 2. Strings as Character Arrays 3. Built-in Functions 4. Common String Problems 5. Activities

1. String Theory
2. **Definition:** A string is a sequence of characters terminated by a null character '\0'.
3. **Characteristics:**
4. Stored as a character array in C.
5. Fixed size if declared as array, dynamic if using pointers.
6. **Advantages:**
7. Can store textual data.
8. Easy manipulation using character arrays.
9. **Disadvantages:**
10. C does not provide built-in dynamic resizing.
11. Manual handling of null-terminator is needed.
12. **Applications:**
13. Processing text, user input, data parsing, file handling.

14. Strings as Character Arrays

15. Declaration:

```
char str[20]; // can hold up to 19 characters + null character
char str2[] = "Hello";
```

16. Accessing Characters:

```
printf("%c", str[0]); // prints first character
str[1] = 'a';
```

17. Built-in Functions

18. **strlen:** Returns length of string (excluding '\0')

```
int len = strlen(str);
```

19. **strcpy:** Copies one string to another

```
strcpy(dest, src);
```

20. **strcmp:** Compares two strings

```
int cmp = strcmp(str1, str2); // 0 if equal
```

21. **strcat:** Concatenates strings

```
strcat(str1, str2);
```

22. Common String Problems

23. Reverse String

```
void reverseString(char str[]){  
    int n = strlen(str);  
    for(int i=0;i<n/2;i++){  
        char temp=str[i]; str[i]=str[n-1-i]; str[n-1-i]=temp;  
    }  
}
```

24. Remove Outermost Parentheses

```
// Stack-based approach can be used, or counting open/close parentheses
```

25. Largest Odd Number in String

```
// Iterate from end, check for odd digit, substring till that index
```

26. Length of Last Word

```
int lengthOfLastWord(char *s){  
    int len=0, i=strlen(s)-1;  
    while(i>=0 && s[i]==' ') i--;  
    while(i>=0 && s[i]!=' '){ len++; i--; }  
    return len;  
}
```

27. Length of String (GeeksforGeeks)

```
int length(char str[]){  
    int len=0;  
    while(str[len]!='\0') len++;  
}
```

```
    return len;
}
```

28. Count the Number of Consistent Strings

```
// Compare each character with allowed set and count valid strings
```

29. Longest Common Prefix

```
// Compare characters of each string index-wise until mismatch
```

30. Rotate String

```
void rotateString(char str[], int k){
    int n=strlen(str);
    k%=n;
    char temp[k];
    for(int i=0;i<k;i++) temp[i]=str[i];
    for(int i=0;i<n-k;i++) str[i]=str[i+k];
    for(int i=0;i<k;i++) str[n-k+i]=temp[i];
}
```

31. Activities **Program: Copy one string to another without using strcpy**

```
#include <stdio.h>
int main(){
    char src[] = "Hello World";
    char dest[50];
    int i;
    for(i=0; src[i]!='\0'; i++) dest[i]=src[i];
    dest[i]='\0';
    printf("Copied string: %s\n", dest);
    return 0;
}
```